Direct or Indirect Manipulation? Evaluating Controller and Virtual Hand Interaction for Drawing and Annotation in Virtual Reality

TRISTAN HOGAN, Colorado State University, USA

Drawing in Virtual Environments can be done using a variety of interaction techniques. This paper compares the differences in speed and accuracy between two common interaction techniques: controller ray-casting and virtual hands. The virtual hands use virtual objects in the environment to draw and the controllers use the ray casting and button presses to draw. This paper also examined the effect of drawing surface size in terms of speed and accuracy. Through a single participant pilot study, controllers had averaged higher accuracy and speed scores. The size of the drawing surface showed that a smaller size increases speed, but decreases accuracy. These results contribute towards our current level of hand tracking capabilities and what software implementations can improve drawing in virtual reality.

Additional Key Words and Phrases: Virtual Reality, Virtual Environment, Virtual Hands, Ray-Casting, Direct Manipulation, Indirect Manipulation

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1 Introduction

For simple tasks such as drawing and writing in virtual environments (VE), finding out whether controller or natural based interaction techniques are more effective, has remained a challenge. While controller based interaction provides higher accuracy and ease of use, natural interactions such as hand tracking can provide a more immersive and realistic experience [4]. With recent improvements in hand tracking combined with the use of virtual hands, this interaction technique has become a more viable option for interacting within VEs.

This paper will compare controller based indirect manipulation, and virtual hand based direct manipulation to explore which technique has greater performance, in terms of speed and accuracy. The controller utilizes the ray cast functionality in combination with physical triggers on the controller to draw within the VE. The virtual hands use virtual objects as physical proxies to draw within the VE. While prior work has examined ray-casting for basic object interaction and virtual hand based approaches for drawing within VEs[6][8][4][2], few studies have directly compared these interactions within the context of drawing and written annotation. Both techniques have unique issues associated with them. Hand tracking capabilities suffer from high latency and inconsistencies with gesture detection[4][8]. Controllers lower the overall immersion by requiring physical objects to be held and lack the grip similarity from drawing in real life[4][8].

This paper aims to investigate the following questions:

- Does controller based indirect manipulation or virtual hand based direct manipulation affect annotation speed and accuracy within a VE?
- Does the size of the drawing surface noticeably affect drawing speed and accuracy?

Author's Contact Information: Tristan Hogan, Colorado State University, Fort Collins, Colorado, USA, Tristan.Hogan@colostate.edu.

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I hypothesize that indirect manipulation will have greater speed, but direct manipulation will have greater accuracy. This is because direct manipulation better follows drawing in real life. However, indirect manipulation does not encounter the same tracking inconsistencies meaning the overall speed will be greater. This paper makes the following contributions:

- A VE with indirect and direct drawing capabilities on two main objects including a whiteboard and sticky notes.
- A pilot study of direct comparison between indirect and direct annotation focusing on speed and accuracy.
- A comparison between small and large drawing surfaces seeing if they affect accuracy or speed.

2 Related Work

2.1 Direct and Indirect Interaction Techniques

Creating immersive interaction techniques in virtual reality (VR) is a large part of what makes a VR immersive. Interactions have two main components: selection and manipulation [1][4]. Selection covers the act of choosing an object to be used for a purpose[1]. Manipulation covers the act of altering a selected object's features or a means to accomplish tasks within the VE [1]. To select and manipulate objects, VR systems use direct and indirect interaction techniques to accomplish this. [4]. Indirect manipulation involves the use of a physical real world object like a controller to interact with objects in a VE [4]. Direct manipulation involves the use of our physical bodies and appendages in tandem with tracking software to directly interact with VEs [4][6]. As research has been done on both manipulation types, multiple studies have found that direct manipulation implementations have lower accuracy and lower user satisfaction compared to indirect manipulation[4][8][2][6]. However, they disagree on whether improving hardware can narrow the gap between them[4][8]. Indirect manipulation has some differences between implementations. Some studies found that more focused controller designs such as VR pens can further increase accuracy for drawing specific tasks[8][2].

2.2 Ray-Based Controller Interactions

Ray-based controller interactions involve a few core components. A light ray is rendered from the virtual controllers and can be used to hover over objects. This is to help aid where users are pointing to within VEs and to make interacting with virtual objects easier [1]. This ray can be used with controller triggers to select objects and manipulate them within the VE [1]. Some studies found that ray-casting is better for selection tasks, but lacks advanced manipulation other controller interactions can provide[1][6]. While This interaction is one of the most popular techniques due to its ease of use and consistency, studies found that its basic structure can make it less desirable for specific tasks [4][2][6]. However, explicitly evaluating ray-casting for annotation in VR lacks research.

2.3 Virtual Hand Interactions

Virtual Hand interactions are generally more complex in nature compared to other interaction types[4]. Hand tracking hardware and software have to work together to achieve an accurate representation of real life hand motions. The most accessible way hand tracking is accomplished is through cameras on VR headsets [8]. Other ways involve using data gloves which are wearable gloves with sensors used to track hand movements and gestures[8]. This interaction method was popular in early VR research due to the natural feel of the controls[6]. Recently, this method has been picking up traction due to hardware and software advancements in hand tracking [4]. As mentioned before, the ease of use and consistency of hand tracking limit its potential. Despite these findings, comparing virtual hands to ray casting for annotation has not been fully explored.

2.4 3D Drawing

Drawing and annotating in VR is an upcoming and diverse part of the VR landscape. Due to the simplicity of drawing and annotating, many applications have been developed revolving around unique VR specific aspects. For example, 3D drawing allows for users to create brush strokes in a 3D space instead of a 2D plane[7][2]. Others incorporate the use of physical objects to increase immersion and increase consistency[5].

3D sketching allows for a new level of artistic innovation and expression. By using a 3D space for drawing, this can better utilize the power VE can provide. The main input methods that have been explored for 3D sketching are controller based and virtual hand based interactions. implementations[7][2]. Physical controllers and VR pens have superior precision compared to virtual hands[2]. Although, only supporting these interactions is not recommended as some artists would prefer the ability to use more natural interactions[7].

2.5 2D Drawing

Due to the increased precision physical controllers can give, others have researched the benefits of using different controller types within 2D drawing applications. The use of physical styluses for drawing has shown potential use cases [5][2][8]. Styluses show an increase in precision and user enjoyment due to the physical hardware and the interaction feeling more natural compared to other controller methods [8][2]. Combining the physical stylus with a physically aligned virtual surface has shown to further increase user satisfaction and ease of use [5]. However, some argue that the amount of gear required to achieve these results is not realistic for everyone [2][4].

2.6 2D Annotation

For annotation, most of the drawing interaction techniques can be applied and new technologies can also be used. Since writing benefits more from increased precision than immersion, using technologies to further increase that precision is vital. Mid-air writing using handwriting recognition is promising. This allows users to speed up their writing within VR as mistakes are automatically corrected during writing[3]. Although, non-drawing techniques such as virtual keyboards are also viable for annotation within VR as they are equally or slightly better than handwriting recognition [3]. Virtual Keyboards also have a smaller learning curve than mid-air writing. For users with marginal or no experience, virtual keyboards have a higher average words per minute than mid-air writing[3].

With these topics in mind, this paper plans to explore the areas of research that are lacking. A direct comparison of indirect and direct manipulation interaction methods in terms of annotation and surface size has not been thoroughly researched. By finding the strengths and weaknesses of both proposed interactions, our understanding of these techniques will continue to grow. With the added variable of drawing surface size, this will further exacerbate any strengths or weaknesses present in these techniques.

3 Methodology

3.1 Equipment

For software, the VR program was developed in Unity version 6000.3.10f1 using the XR Interaction Toolkit version 3.3.1 and XR Hands version 1.7.3. The program is an immersive VE with a drawable whiteboard, drawable sticky notes and virtual marker objects for virtual hand drawing. Both the whiteboard and sticky notes can be drawn on using the controllers and virtual hands. The controllers use the XR NearFarInteractor to raycast onto the drawable surface in combination with trigger presses to draw. For virtual hands, there are two marker objects within the VE that can be picked up and used as a normal marker. To draw with the marker, collide the marker's tip to a drawable surface. For hardware, the program was run on an ASUS ROG Zephyrus G14 laptop. The laptop is equipped with an AMD Ryzen 9 7940HS running at 4 GHz, 32 GB of 4800 MT/s DDR5, and a Nvidia Ge-force RTX 4060 Mobile with 8GB of VRAM running at 2000 MHz. The system was running on Windows 11. The headset used was a Meta Quest 2 with standard controllers.

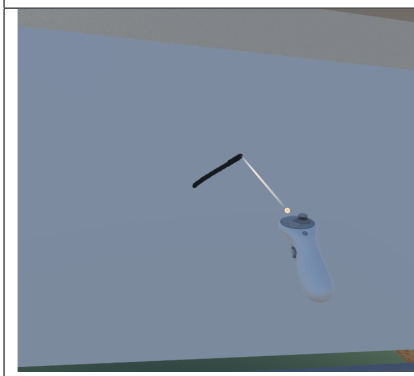
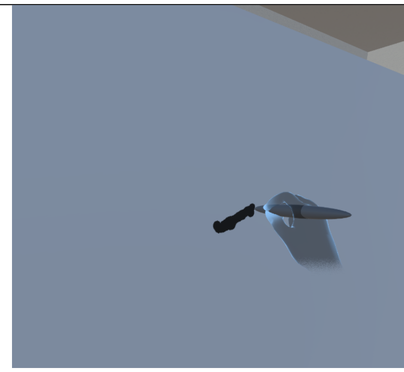
Controllers	Virtual Hands
	

Table 1. Both Interaction Techniques Drawing on the Whiteboard

Controllers	Virtual Hands
	

Table 2. Both Interaction Techniques Drawing on the Sticky Note

3.2 Procedure

Each participant puts on the headset and gets time to familiarize themselves with the environment. They get time to try both interaction methods to understand how the controls work. To begin, the participant will start on the whiteboard surface first, then move onto the sticky note surface. They will complete six trials, alternating from the controller to virtual hands. For each trial, the participant will select the controller or virtual hands teleport pad to set their distance. Then, the participant will try their best to write within the lines of the sentence template on the whiteboard. At the end of each trial, a screenshot of the result is taken for analysis. For the next surface, the participant will do most of the same steps. The whiteboard object will be replaced with a sticky note object, and the participant will only be able to select the teleport pad closest to the sticky note. They will alternate from the controller to virtual hands for each trial. Then, the participant will trace the same sentence trying their best to stay within the lines. After each trial, a screenshot of the result will be taken.

3.3 Design

This study was completed as a single participant pilot study. The experiment is a within-subject design with two independent variables and two dependent variables. The independent variables are; interaction technique and surface type. The dependent variables are; accuracy and speed. Each task has a total of three trials. With four conditions and three trials, each participant completes a total of twelve trials. The tasks were grouped by surface with each surface having alternating techniques for each surface. For the task, each participant is shown a faint template of the sentence "The quick brown fox." This sentence was chosen due to the variety of letters and overall length. To track accuracy, each result is graded on a qualitative scale of 1-5. Speed was calculated by recording each trial and finding the exact start and end times using video editing software. These values were recorded and used at the end to calculate the average speed and accuracy for each surface and interaction type.

Score	Description
1	Horrible accuracy. Less than a quarter of the brush strokes follow the template. At least four major deviations, tons of minor deviations. Letters are nearly illegible.
2	Bad accuracy. Less than half of the brush strokes follow the template. Has lots of minor deviations and at most three major deviations. Letters are difficult to read.
3	Average accuracy. More than half of the brush strokes follow the template, with some minor deviations and at most one major deviation. Letters are somewhat readable.
4	Good accuracy. Most brush strokes follow the template with at most two minor deviations. Letters are readable and closely resemble the template's form.
5	Excellent accuracy. Brush strokes follow the template with at most one minor deviation. Letters are readable and nearly resemble the template perfectly.

Table 3. Accuracy rating scale used for evaluating trial results

Score 1 **Score 2** **Score 3** **Score 4** **Score 5**

Fig. 1. Example results for each accuracy score from 1 to 5

4 Results

As shown in table 4, each trial was timed and rated between 1-5 based on the perceived accuracy of the written words. Then, the average time and score for both surfaces and techniques was calculated at the bottom.

4.1 Task Completion Time

The differences in task completion time can help show which interaction type is easier to use and more user friendly. By calculating the averages, a significant difference is seen. As seen in Table 4, the controller interaction was on average 48% and 39% faster than the virtual hands for the whiteboard and sticky note surfaces respectively. The controllers averaged 43.73 seconds and 26.92 seconds compared to an 83.56 second and 44.34 second average time for the virtual hands. For the whiteboard, the fastest time the controllers achieved was a 38.87 second time. The fastest time virtual hands achieved was a 72.31 second time. For the sticky notes, the fastest time the controllers achieved was a 24.59 second time. The fastest time for the virtual hands was a 35.14 second time.

4.2 Accuracy

Accuracy is a key value that can help determine an interaction's efficiency and consistency. By averaging the scores for each interaction, an important insight can be drawn for both the interactions and drawing surfaces. As seen in Table 2, the controller interaction achieved on average 20% and 133% higher accuracy scores than the virtual hands. For the whiteboard, controllers averaged a score of 4 and the virtual hands averaged a 3.33. For the sticky note, the controllers averaged a score of 2.33 while the virtual hands averaged a score of 1.

4.3 Surface Types

The surface types showed the difference the size of the drawing surface can make. The whiteboard averaged 4 and 3.33 points for accuracy while the sticky note averaged 2.33 and 1 points. The speed also changed significantly. The average controller speed was 38% faster on sticky notes compared to the whiteboard. The average virtual hand speed was 47% faster on sticky notes compared to the whiteboard.

Trial	Surface	Technique	Time (s)	Accuracy (1-5)
1	Whiteboard	Controller	50.3	4
2	Whiteboard	Marker	87.02	3
3	Whiteboard	Controller	42.03	3
4	Whiteboard	Marker	91.35	4
5	Whiteboard	Controller	38.87	5
6	Whiteboard	Marker	72.31	3
7	Sticky Note	Controller	24.59	2
8	Sticky Note	Marker	51.11	1
9	Sticky Note	Controller	26.45	2
10	Sticky Note	Marker	35.14	1
11	Sticky Note	Controller	29.73	3
12	Sticky Note	Marker	46.43	1
Avg	Whiteboard	Controller	43.73	4
Avg	Whiteboard	Marker	83.56	3.33
Avg	Sticky Note	Controller	26.92	2.33
Avg	Sticky Note	Marker	44.23	1

Table 4. Trial results for all 12 trials and condition averages

5 Discussion

These results show a major difference between the interaction techniques and the drawing surfaces. For the interaction techniques, the controllers were faster and more accurate on every trial. The virtual hands never achieved a faster time than the controllers and only got one trial with a higher score than a controller trial. The main reasons behind this difference is due to the drawing implementation and hardware differences. For drawing, the controllers make use of linear interpolation. Linear interpolation is a technique for smoothly transitioning from one position to another. To better explain this, imagine there are two points A and B located on a 2D plane. Without linear interpolation, moving from A to B would be done instantly in one frame. With linear interpolation using a factor of 0.1, the position would move 10% of the remaining distance every frame. This means any small jitters or sudden movements are dampened making drawing feel more consistent and manageable. Since controllers lack stability during use, drawing without linear interpolation makes drawing near impossible and frustrating. The virtual markers did not make use of this technique as during project creation, I did not feel as if they needed it due to the natural stability of pinch gesture and distance to the drawing surfaces. For hardware differences, the virtual hands had issues with maintaining accurate hand tracking and gesture tracking. The virtual hands would occasionally inaccurately track the hands causing the virtual marker to start drawing on incorrect areas. The gesture tracking would cause the virtual hands to stop holding the virtual marker slowing down the task completion times. For the drawing surfaces, size was the main limitation on drawing accuracy. The size of the template on the whiteboard was large enough so small mistakes would not go outside the lines and following the section being drawn on was easier. On the sticky notes, any small mistakes would make the brush go outside the template lowering accuracy and increase time spent correcting. While using the virtual marker, it was difficult to follow exactly where the marker was drawing lowering accuracy and speed even further. This greatly impacted the accuracy and speed of both interactions, but the virtual hands felt the performance impacts more severely.

5.1 Limitations

There are a few noticeable limitations that I found throughout this study. Primarily, the use of linear interpolation on only one of the interaction techniques. I incorrectly assumed that the stability of drawing with hand gestures would not require linear interpolation and that it would not make a large enough difference. The lack of physical feedback from the virtual marker and virtual board compared to drawing in real life decreased the overall stability more than I anticipated causing the usability to suffer. Secondly, the Meta Quest 2 is an older VR headset meaning the latest advances in software and hardware are not seen in this study. Newer models with better hand tracking and gesture tracking could yield better results and show current progress with hand tracking. Lastly, the study design was lacking in a few areas. Since the results were gathered from a single participant, the results are weaker and not generalizable. There was no counter balancing as all of the whiteboard trials were completed before the sticky note trials. The quantitative evaluation of accuracy is subjective and has the potential to introduce bias. Since I was the developer and participant, I have more experience and understand what approach I should take for each interaction and surface.

6 Conclusion

Overall, this paper explored the differences between direct and indirect manipulation for drawing and annotating in an immersive VE. I tested the differences between using physical controllers with ray casting and virtual hands with virtual markers for writing a basic sentence on a whiteboard and sticky notes. The results show that controllers have greater accuracy and speed compared to virtual hands. The size of the drawable surface significantly impacts drawing performance as the sticky notes had noticeably worse accuracy scores compared to the whiteboard. I found that the current state of hand tracking and lack of physical feedback during drawing makes using virtual hands and virtual markers more difficult and less stable than real life. The use of linear interpolation greatly impacts the ease of use for drawing in VR as both interaction techniques caused shakiness in movements. This shakiness drastically increases the difficulty and frustration during drawing, but linear interpolation mitigates this issue. My results did not support my hypothesis as I figured the virtual hands would achieve greater speed due to the grip used to hold the virtual marker being more familiar. However, if the virtual hands technique made use of linear interpolation, the differences between each method could be different. For future work, using more recent VR hardware and making both interactions use or not use linear interpolation could show stronger results. Potentially aligning the virtual drawing surface to a physical surface in real life could increase the usability of virtual hands.

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